
Report 2

Breast-cancer-prediction

Dataset = WISCONSIN DIAGNOSIS BREAST CANCER(WDBC)

Friday, 22th june 2024

Content:

COMPARING 3 CATEGORIES RESULTS:

1.1 **WITHOUT HYPER-TUNING:**

1.1.1 MODEL EVALUATING (*CROSS-VAL, WITHOUT CROSS-VAL*)

1.2 **HYPERTUNING:**

1.2.1 MODEL EVALUATING(*RANDOM-SPLIT, GRID-SEARCH*)

Information:

- Dataset has splitted with a stratified method into : train = 70, test = 15, validation = 15.
 - In Cross-Validation CV = 6 for all algorithms was used.
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Algorithms	SVM	RF	KNN
Stratified-Split			
Without CV	92% 94%	94% 97%	94% 94%
Cross-Validation	92% 90%	94% 96%	93% 92%

- Using cross-validation in this dataset, not only had significant improvement , but also decreased the accuracy of models.

Algorithms	SVM	RF	KNN
Grid-Search			
Without CV	97% 97% ▲	93% 95% ▲	90% 94% ▲
Cross-Validation	94% 95%	93% 96% ▲	94% 90%

- SVM : {'C': 1, 'gamma': 'scale', 'kernel': 'linear'}
- RF : {'bootstrap': True, 'criterion': 'entropy', 'max_depth': 15, 'max_features': 0.5, 'min_samples_leaf': 8, 'n_estimators': 40}
- KNN : {'n_neighbors': 5}

▲ $\text{increase}(\text{test}(\text{cv}) / \text{test}(\text{without cv}))$

Algorithms	SVM	RF	KNN
Random-Search			
Without CV	97% VAL ACCURACY TEST 95%	92% VAL ACCURACY TEST 95%	94% VAL ACCURACY TEST 93%
Cross-Validation	95% VAL ACCURACY TEST 96% ▲	93% VAL ACCURACY TEST 96% ▲	93% VAL ACCURACY TEST 92%

- SVM : {'C': 7.631006430303977, 'kernel': 'linear'}
- RF : {'criterion': 'gini', 'max_depth': 42, 'max_features': 35, 'min_samples_leaf': 5, 'min_samples_split': 3, 'n_estimators': 67}
- KNN : {'n_neighbors': 5}

Algorithms	Highest score
SVM	97%(Grid-Search, without CV)
RF	97%(Stratified Split method)
KNN	94%(Grid-Search, without CV)

- Using CV for model accuracy was not effective, except in 1)RF in the Grid-search and 2)RF in the Random-Search methods and 3)SVM in the Random-Search method.
- Using hyper-tuning for model accuracy was effective only for 1)SVM in the Grid-Search.
- Grid-Search was more effective for model accuracy than Random-Search.
- However, using CV in Random-Search could enhance the model accuracy more than the CV in Grid-Search.